```
module add_sub_4bit (
    input  [3:0] a,
    input  [3:0] b,
    input        add_nsub,
    output reg [3:0] sum,
    output reg       carry_out
);

    reg [4:0] r_result;

    always @(*) begin
        if (add_nsub == 1'b1) begin
            r_result = a + b;
        end
        else begin
            r_result = a - b;
        end
    end

    always @(*) begin
        carry_out = r_result[4];
        sum       = r_result[3:0];
    end

endmodule
```

```v
module tb_add_sub_4bit();

    // 입력 신호
    reg [3:0] a;
    reg [3:0] b;
    reg       add_nsub;

    // 출력 신호
    wire [3:0] sum;
    wire       carry_out;

    add_sub_4bit u_add_sub (
        .a(a),
        .b(b),
        .add_nsub(add_nsub),
        .sum(sum),
        .carry_out(carry_out)
    );

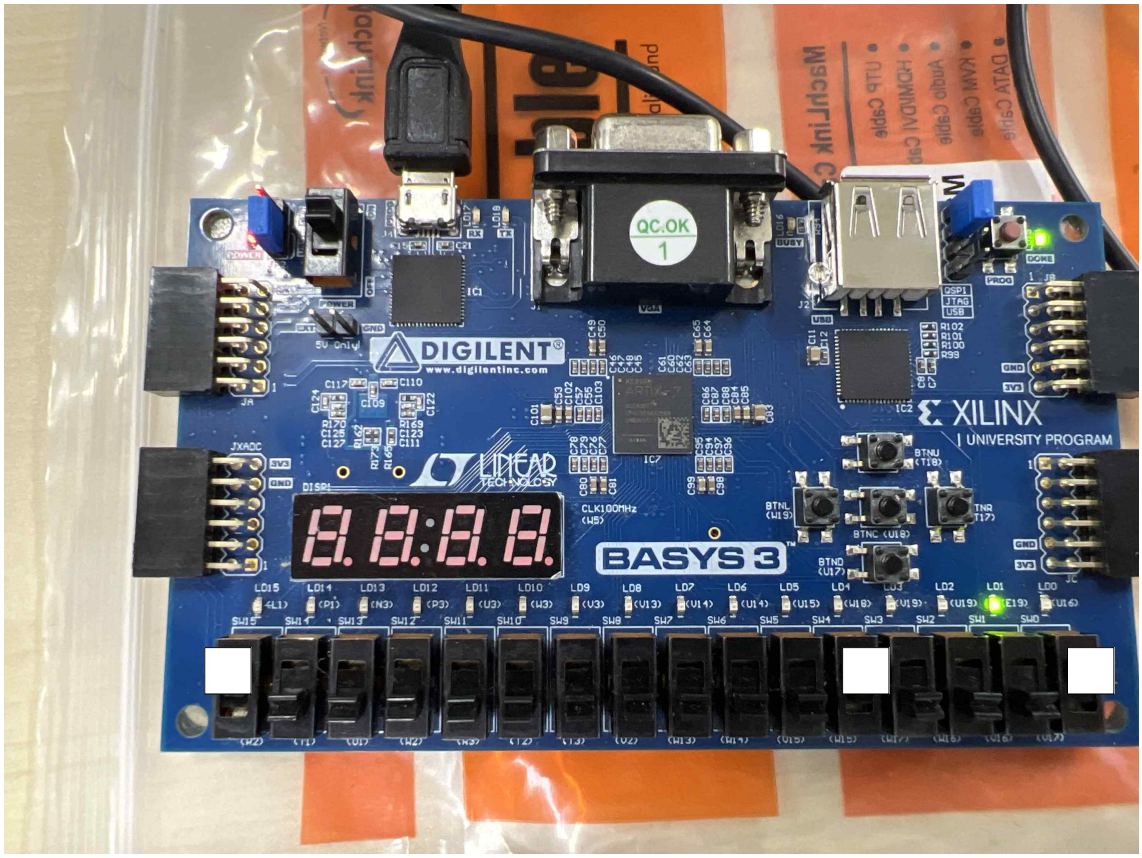
    initial begin
        // 초기값 설정
        a        = 4'b0000;
        b        = 4'b0000;
        add_nsub = 1'b1; // 기본 가산 모드

        #10; a = 4'b0101; b = 4'b0011; add_nsub = 1'b1;
        #10; a = 4'b1111; b = 4'b0001; add_nsub = 1'b1;
        #10; a = 4'b1010; b = 4'b0101; add_nsub = 1'b1;

        #10; a = 4'b1010; b = 4'b0011; add_nsub = 1'b0;
        #10; a = 4'b0101; b = 4'b1000; add_nsub = 1'b0;
        #10; a = 4'b1111; b = 4'b1111; add_nsub = 1'b0;

        #10;
        $finish;
    end

endmodule
```


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Untitled 1*
tb_add_sub_4bit.v
Name
Value
a[3:0]
0000
b[3:0]
0000
add_nsub
1
sum[3:0]
0000
carry_out
0
0.000 ns
10.000 ns
20.000 ns
30.000 ns
40.000 ns
50.000 ns
60.000 ns
0000
0101
1111
1010
0101
1111
0000
0011
0001
0101
0011
1000
1111
0000
1000
0000
1111
0111
1101
0000